Daizhi Liao

Undergraduate researcher · Reinforcement Learning & Embodied AI · PhD applicant Fall 2027

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EDUCATION

Guangzhou University — B.Eng., Network Engineering

Expected 2027

Fourth-year undergraduate (senior; enrolled 2023, expected graduation 2027). Focus: reinforcement learning, embodied AI, and the empirical auditing of deployed AI systems.

RESEARCH & PROJECTS

wm-audit — Preregistered Audit of World-Model Fidelity vs Control Utility

2026

Single-machine CPU audit suite (reproducible, seeded) that separates world-model *perceptual fidelity* (next-frame MSE / SSIM / NLL; fixed random-conv filter banks as stand-ins for learned metrics) from *control utility* (whether a planner can solve the task). Reduced pilot on one env (pixel_nav, 3 seeds × 3 capacities, 12k steps): H1 threshold met — pixel fidelity weakly tracked closed-loop return at $|\rho|=0.32<0.4$, but this is a weak, tentative exploratory signal only (n=9; 95% CI crosses zero, Holm-adj p=1.0 → not statistically significant, not an established finding); control-relevant diagnostics did not (H2 not supported, $\rho=0.26<0.6$); INT8 compression did not degrade calibration faster than raw fidelity (H3 not supported). The locked 200k-step × 5-seed × 2-env main run has not been run; this is the preregistered reduced pilot only.
Code: github.com/dafahaha/wm-audit

TransitTruth — Behavioral-Fingerprint Audit of AI API Relays

2026

Solo design & implementation. Open-source tool that verifies whether an AI API relay actually serves the advertised model, by probing statistical behavior (random-number, token, latency distributions) against baselines with chi-square / KS tests and Bayesian updating. A guided offline demo accompanies the tool (run the real audit locally with `transit-truth serve`); a technical report validates same-family discrimination: gpt-4o vs gpt-4o-mini, TVD up to 0.90, 91.3% in-distribution resubstitution posterior (not held-out).
Code & demo: github.com/dafahaha/transit-truth · live demo

embodied-offpolicy-study — Pre-registered Two-Arm SAC Study with Logged Third-Arm Amendment on Hopper-v4

2026

Soft Actor-Critic implemented from scratch in PyTorch (no Stable-Baselines3), run as a pre-registered two-arm controlled study with a logged third-arm (FixedAlpha) amendment on Hopper-v4 (CPU, 120k env steps, 3 seeds/arm): Baseline (reward scale 1.0) vs RewardScaled (0.1) vs FixedAlpha (scale=0.1, α fixed at 0.2; automatic temperature off). Mean return ± SD (final window 105k–120k, n=3/arm): Baseline 1085±698, RewardScaled 1118±623, FixedAlpha 1404±632. Welch p: Baseline–FixedAlpha 0.59, Baseline–RewardScaled 0.95, FixedAlpha–RewardScaled 0.61; one-way ANOVA F=0.22, p=0.81 — no statistically distinguishable effect detected (n=3; the t-intervals overlap heavily). At 60k the reward-scaled arm looked ~3× better, but extending to the pre-registered 120k budget and adding the fixed-temperature control overturned that apparent gain as early seed noise: an honest pre-registered null. Raw CSVs committed, curves reproduced by script.
Code: github.com/dafahaha/embodied-offpolicy-study

rl-deploy-bench — Cross-Platform RL Deployment & Benchmarking

v1.1.0

Config-driven CLI: export Stable-Baselines3 policies to ONNX/TorchScript, quantize to FP16/INT8, and benchmark latency, throughput, and accuracy across x86 GPU and CPU via ONNX Runtime, with auto-generated HTML reports. An experimental standalone TensorRT engine-builder (FP16/INT8) is included but not yet wired into the benchmark harness (GPU-tested numbers tracked for v2.0). Jetson/edge is a target deployment path, not yet measured on device. A reproducible harness for the accuracy–latency tradeoff of quantized RL policies.

PUBLICATIONS & PREPRINTS

- Behavioral Fingerprinting of LLM API Relays — detecting model substitution on AI API relays (double-blind workshop paper). *In preparation.*
- TransitTruth: Extended Technical Report — behavioral probing, chi-square/KS & Bayesian self-consistency, threats to validity (2026). PDF.

OPEN SOURCE — MERGED CONTRIBUTIONS (10)

Code fixes: microsoft/TextWorld #377 — fixed `env.step` crash on multi-command input (non-greedy score parsing + regression test); **shmuma/ptan #57** — fixed Gymnasium 5-tuple API compatibility in experience sources; **learnsyslab/gym-pybullet-drones #319** — converted `print()+exit()` error paths to raised exceptions across 8 files; **MichalBortkiewicz/JaxGCRL #65** — fixed XML path resolution across all arm manipulation environments; **RLE-Bench/RLE-Bench #9** — recorded per-trial status/failure reason/timing in `summary.json` and fixed embedded final-step reward resolution.

Community & documentation (not code): Eclectic-Sheep/sheeprl #340 (CONTRIBUTING, Code of Conduct); Algorineko/AgenticArXiv-RL #72 (MIT LICENSE, CONTRIBUTING, CoC, CI); FluxVLA #122 (`bloat16→bf16` typo) and #123 (community docs); scadastrangelove/awesome-ai-security-tools #165 (WATCHLIST entry, 2026-10-21).

Beyond merged work, additional code PRs are **currently under review (not yet merged)** upstream in e.g. vLLM, pandas, google/brax, Farama-Foundation/PettingZoo, thu-ml/tianshou, ContinualAI/avalanche, and Hugging Face datasets & diffusers.

TECHNICAL SKILLS

RL & learning: PyTorch, Stable-Baselines3, Gymnasium, MuJoCo | **Deployment:** ONNX, ONNX Runtime, TorchScript, INT8/FP16 quantization; TensorRT (experimental, not yet benchmarked) | **Systems:** Python, Linux, Docker, Git, CI/CD; CUDA (learning, no GPU-measured results yet)